Database System Engineering and Distributed Backend Development
Course Code: 25CS1302E

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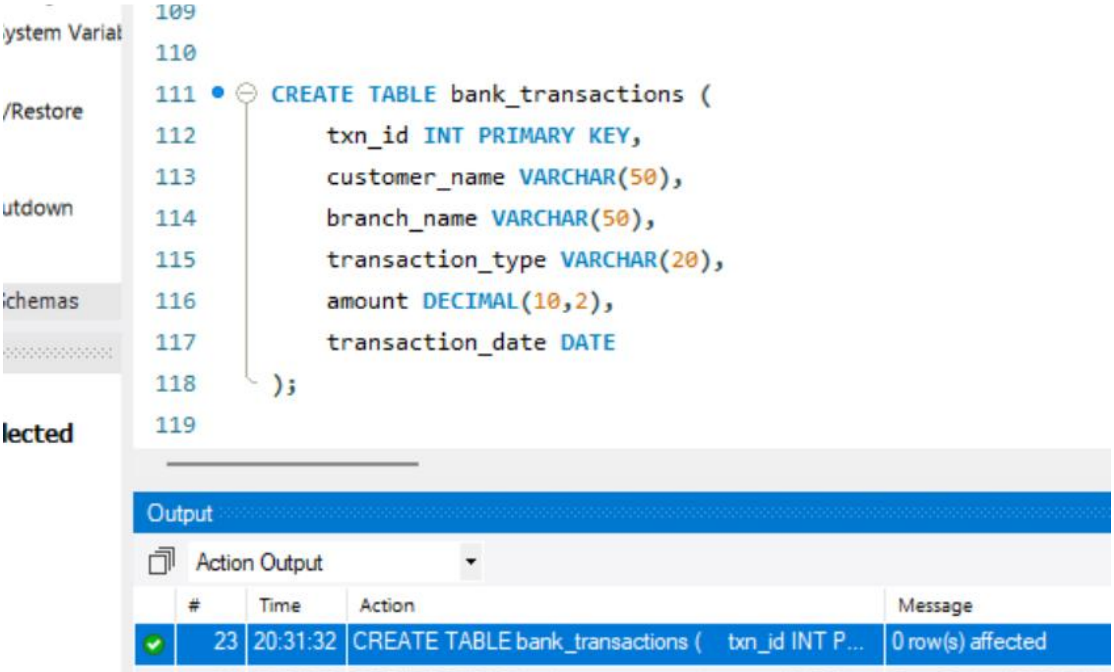
STEP1: CREATE TABLE



```
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```


STEP 2: INSERT DATA

```sql
INSERT INTO bank_transactions VALUES
(101,'VAISHNAVI','Hyderabad','Deposit',5000,'2024-01-05'),
(102,'LIKHITHA','Hyderabad','Withdrawal',2000,'2024-01-06'),
(103,'SUSHMA','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'AKHILA','Vizag','Deposit',8000,'2024-01-10'),
(105,'DIVYA','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'DISKHITHA','Vizag','Deposit',15000,'2024-01-12'),
(107,'ARCHANA','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'BABY','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'SAI','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'KUSHAL','Vijayawada','Deposit',11000,'2024-01-16');
```

Output

Action Output

	#	Time	Action	Message
✖	24	20:31:35	CREATE TABLE bank_transactions (txn_id INT P...	Error Code: 1050. Table 'bank_transactions'
✔	25	20:33:37	INSERT INTO bank_transactions VALUES (101,'VAI...	10 row(s) affected Records: 10 Duplicates:

STEP 3: SUM ()

```sql
SELECT SUM(amount) AS Total_Amount
FROM bank_transactions;
```

Result Grid | Filter Rows: | Export:

Total_Amount
70500.00

Step 4: AVG ()

```sql
SELECT AVG(amount) AS Average_Transaction
FROM bank_transactions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Average_Transaction
7050.000000

STEP 5: MAX ()

```sql
SELECT MAX(amount) AS Highest_Transaction
FROM bank_transactions;
```

Highest_Transaction
15000.00

STEP : MIN ()

```sql
SELECT MIN(amount) AS Lowest_Transaction
FROM bank_transactions;
```

Lowest_Transaction
1000.00

STEP 7 : COUNT()

```sql
SELECT COUNT(*) AS Total_Transactions
FROM bank_transactions;
```

Total_Transactions
10

STEP 8 : WHERE + SUM()

```sql
SELECT SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Result Grid | Filter Rows: | Export:

Total_Deposit
60000.00

STEP 9 : GROUP BY

```sql
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Result Grid | Filter Rows: | Export:

branch_name	Total_Amount
Hyderabad	19500.00
Vijayawada	24000.00
Vizag	27000.00

STEP 10 : GROUP BY + HAVING

```sql
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING SUM(amount) > 20000;
```

Result Grid | Filter Rows: | Export:

branch_name	Total_Amount
Vijayawada	24000.00
Vizag	27000.00

STEP 11: GROUP BY + ORDER BY

```sql
SELECT branch_name,
        SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
ORDER BY Total_Amount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap

branch_name	Total_Amount
Vizag	27000.00
Vijayawada	24000.00
Hyderabad	19500.00

STEP 12 : GROUP BY + HAVING + ORDER BY

```sql
SELECT branch_name,
        COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) >= 3
ORDER BY Total_Transactions DESC;
```

Result Grid | Filter Rows: | Export:

branch_name	Total_Transactions
Hyderabad	4
Vijayawada	3
Vizag	3

STEP 13: WHERE + COUNT ()

```sql
SELECT COUNT(*) AS Withdrawals
FROM bank_transactions
WHERE transaction_type='Withdrawal';
```

Result Grid | Filter Rows: | Export:

Withdrawals
4

STEP 14: WHERE + AVG ()

```sql
SELECT AVG(amount) AS Avg_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Result Grid Filter Rows: Export:

Avg_Deposit
10000.000000

STEP 15: GROUP BY + MAX()

```sql
SELECT branch_name,
        MAX(amount) AS Highest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Result Grid Filter Rows: Export:

branch_name	Highest_Amount
Hyderabad	9000.00
Vijayawada	12000.00
Vizag	15000.00

STEP 1: GROUP BY + MIN ()

```sql
SELECT branch_name,
        MIN(amount) AS Lowest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Result Grid Filter Rows: Expo

branch_name	Lowest_Amount
Hyderabad	2000.00
Vijayawada	1000.00
Vizag	4000.00

Step 17: WHERE + ORDER BY

```sql
SELECT *
FROM bank_transactions
WHERE amount > 8000
ORDER BY amount DESC;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	DISKHITHA	Vizag	Deposit	5000.00	2024-01-12
103	SUSHMA	Vijayawada	Deposit	12000.00	2024-01-08
110	KUSHAL	Vijayawada	Deposit	11000.00	2024-01-16
108	BABY	Hyderabad	Deposit	9000.00	2024-01-14

Step 18 : GROUP BY + COUNT ()

```sql
SELECT branch_name,
       COUNT(*) AS Customer_Count
FROM bank_transactions
GROUP BY branch_name;
```

branch_name	Customer_Count
Hyderabad	4
Vijayawada	3
Vizag	3

STEP 19 : HAVING + COUNT()

```sql
SELECT branch_name,
       COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) > 3;
```

branch_name	Total_Transactions
Hyderabad	4

STEP 20: GROUP BY + ANG ()

```sql
HAVING COUNT(*) > 3;
SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name;
```

branch_name	Avg_Amount
Hyderabad	4875.000000
Vijayawada	8000.000000
Vizag	9000.000000

STEP 21: WHERE + GROUP BY + HAVING

```sql
SELECT branch_name,
       SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit'
GROUP BY branch_name
HAVING SUM(amount) > 15000;
```

branch_name	Total_Deposit
Vijayawada	23000.00
Vizag	23000.00

STEP 22: GROUP BY + HAVING + ORDER BY

```sql
SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING AVG(amount) > 7000
ORDER BY Avg_Amount DESC;
```

branch_name	Avg_Amount
Vizag	9000.000000
Vijayawada	8000.000000

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